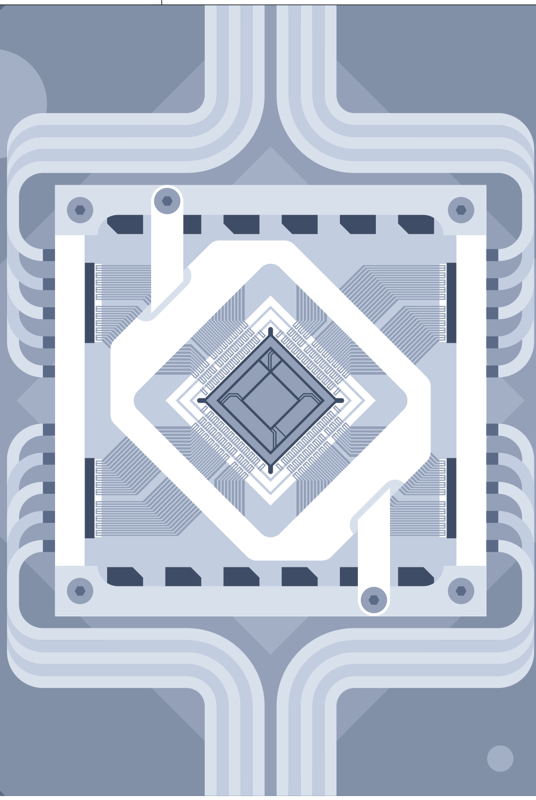
Bees understand nothing

Scientists have uncovered that honeybees can actually grasp the concept of nothing, that most animals cannot. **https://dgit.in/Jul18-76-1**

Bowel-listener belt

Barry Marshall is developing a belt that listens to the sounds of the bowel and diagnoses it **https://dgit.in/Jul18-76-2**



LASER, MEET QUANTUM COMPUTING

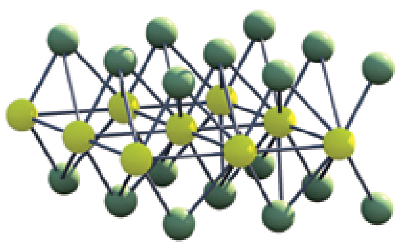
Quantum computing could get a massive upgrade, thanks to super fast lasers

Arnab Mukherjee | arnab@digit.in

Quantum computing, and the work towards realising it, has been around for ages now. Rather than boring you with the milestones that got it where it is today, it suffices to say that it is here. As Isaac Chuang, a lean, soft-spoken MIT professor, puts it "Quantum computing is no longer a physicist's dream, it is an engineer's nightmare". We've had a slew of announcements from companies such as IBM, Google and Intel, each touting their own success, at whatever scale, into breaking the next barrier in quantum computing. Yet, at a macro level, it is evident that we are far from 'quantum supremacy' in practicality. Quantum supremacy, in this context, refers to the situation where quantum computers can actually achieve persistent results faster than a conventional computer. If the findings of a recent research paper are to be believed though, things could turn around pretty soon.

WHAT IT ACHIEVED

A computer works in terms of operations on 1s and 0s. A typical computer in 2018 can perform 1 billion of these operations per second. Thanks to the efforts of a group of researchers from Germany and the USA, computing in the near future might leave that number in the dust.



A lattice representation of WSe2 (Tungsten diselenide)

The experiment conducted involves pulsing infrared light on honeycomb-shaped lattices of tungsten and selenium. This allows the silicon bits to switch between 1 and 0 pretty much like a normal computer - only a million times faster. How does that happen? For that, we need to understand how electrons behave in a honeycomb lattice.

THE PRINCIPLE

When it comes to molecules, electrons are usually in orbit around them in fixed quantum states. They can jump between these states/orbits/tracks as and when they gain or lose energy. These tracks are known as valleys by the researchers and the manipulation of these states as 'valleytronics'.

In the tungsten-selenium lattice, only two tracks are available for electrons to burn off their excess energy. So, when you direct infrared light in a particular orientation at the lattice, you'll excite an electron to move to the first track. Infrared light in a different orientation will excite the electron to move onto the second track. Now, when you think of these tracks as the states of a bit, i.e 1 and 0, you can see how this process can be used for computing.

But using them for this purpose isn't exactly easy. That is because these electrons don't really like hanging around on these tracks for too long. And they don't need to. It only takes them a few femtoseconds to lose the excess energy and return back to the fixed tracks closer to the nucleus. To